# Assignment 2 Report
# Convolution

## Introduction

The purpose of this assignment was to apply convolutional neural networks (convnets) to image classification and to examine how the training sample size affects the choice of picking between a pretrained model or a model developed from scratch. The image classification used in this assignment was the Cats & Dogs example from the course material.

There were two main approaches in this assignment, one being the model from scratch and the other being the pretrained model. Since small datasets are prone to overfitting techniques such as data augmentation, dropout, and callbacks were used to help stop overfitting and improve performance of the models.

The main objective of this assignment was to determine how model performance changed as training sample size increased and to explain the relationship between training sample size and choice of which network.

## Data and Test Setup

The dataset used in the assignment contains images of cats and dogs. For all the tests done the validation and test sample size were kept at a constant 500 images each. The training sample sizes were verified depending on the test.

For the network trained from scratch there were 3 test done:

Test 1: Training Sample Size of 1000
Test 2: Training Sample Size of 1500
Test 3: Training Sample Size of 2000

For the pretrained network there were also 3 test done:

Test 4: Training Sample Size of 1000
Test 5: Training Sample Size of 1500
Test 6: Training Sample Size of 2000

The images were resized up to 180 x 180 pixels. The same validation and test sets were used across all the tests done.

## Methodology

### Network trained from scratch

For the first set of tests, a convolution neural network was built from scratch. The model included multiple convolution and max-pooling layers, followed by a flattening layer, dropout, and a final dense output layer with sigmoid activation and binary classification.To reduce overfitting and improve overall performance the following techniques were used.

- Data augmentation
- Dropout regularization
- Callbacks
- Early stopping

The model was compiled using binary crossentropy loss, the RMSprop optimizer, and accuracy as the evaluation metric.

### Pretrained Network

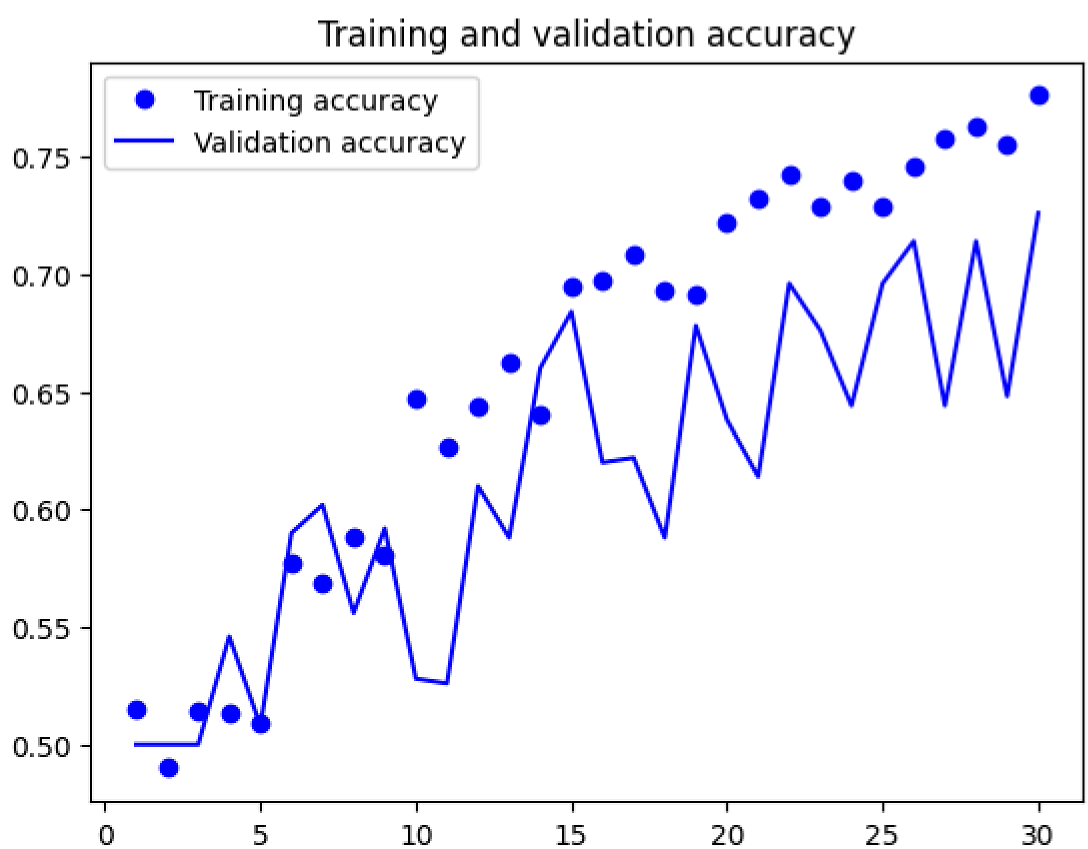
For the second set of tests, a pretrained convolution neural network was used. In this section of the tests VGG16 was used as a feature extractor. Additional dense layers were added on top of the pretrained base for binary classification. To improve performance these optimization techniques were used.

- Data augmentation
- Freezing the early layers
- Fine-tuning selected upper layers
- Model checkpointing
- Early stopping

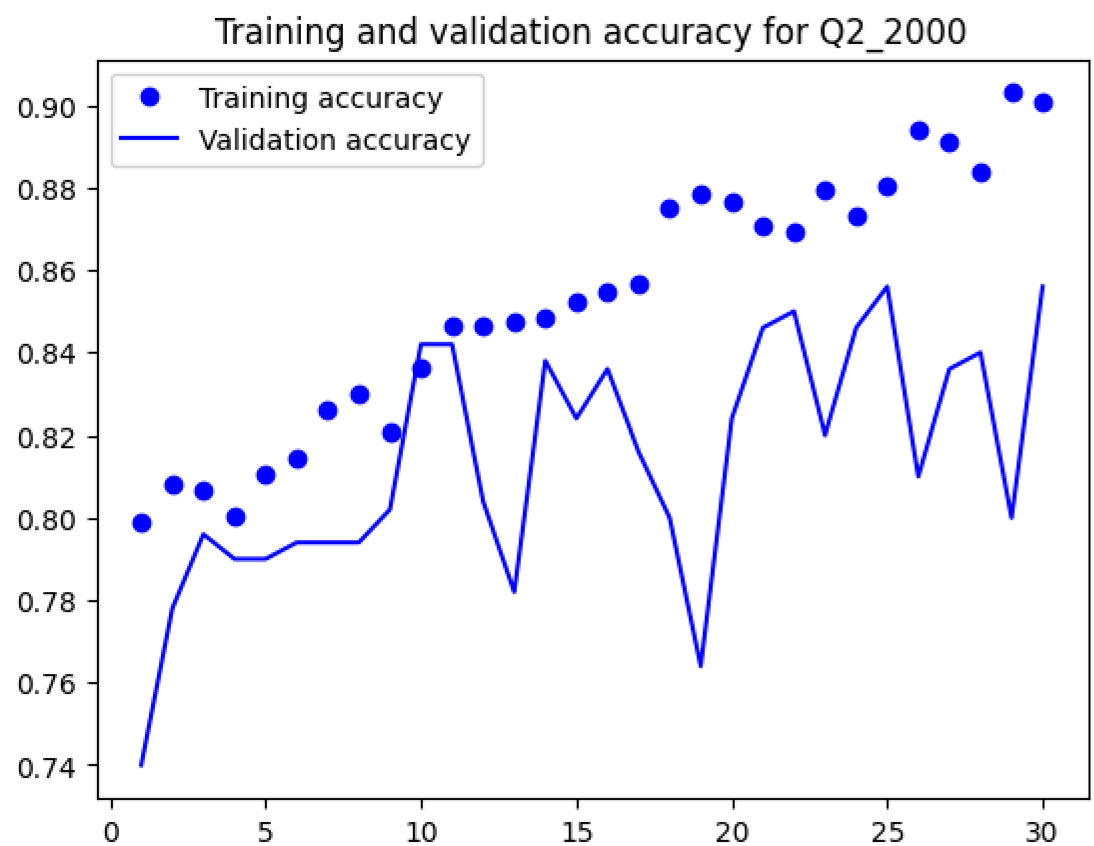
This model was also evaluated using test accuracy.

### Results

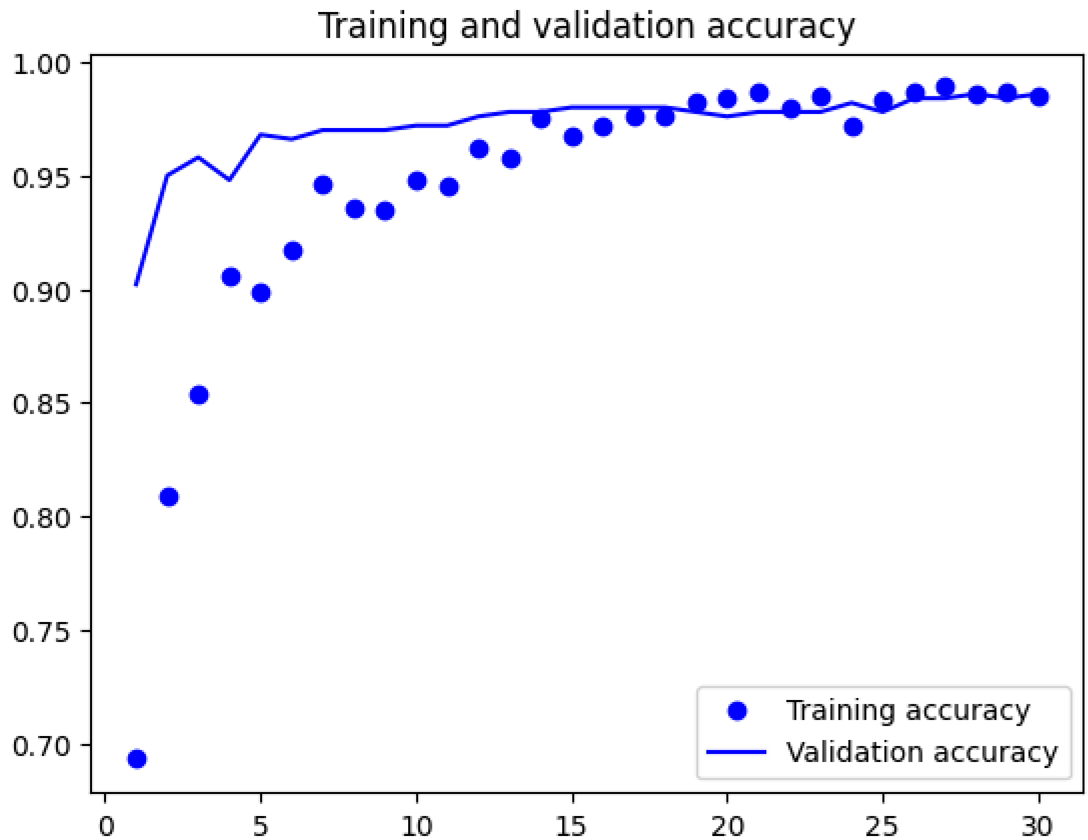
| Test | Network Type | Training Size | Validation Size | Test Size | Test Accuracy |
|---|---|---|---|---|---|
| 1 | Scratch CNN | 1000 | 500 | 500 | .708 |
| 2 | Scratch CNN | 1500 | 500 | 500 | .754 |
| 3 | Scratch CNN | 2000 | 500 | 500 | .828 |
| 4 | Pretrained CNN | 1000 | 500 | 500 | .958 |
| 5 | Pretrained CNN | 1500 | 500 | 500 | .982 |
| 6 | Pretrained CNN | 2000 | 500 | 500 | .974 |



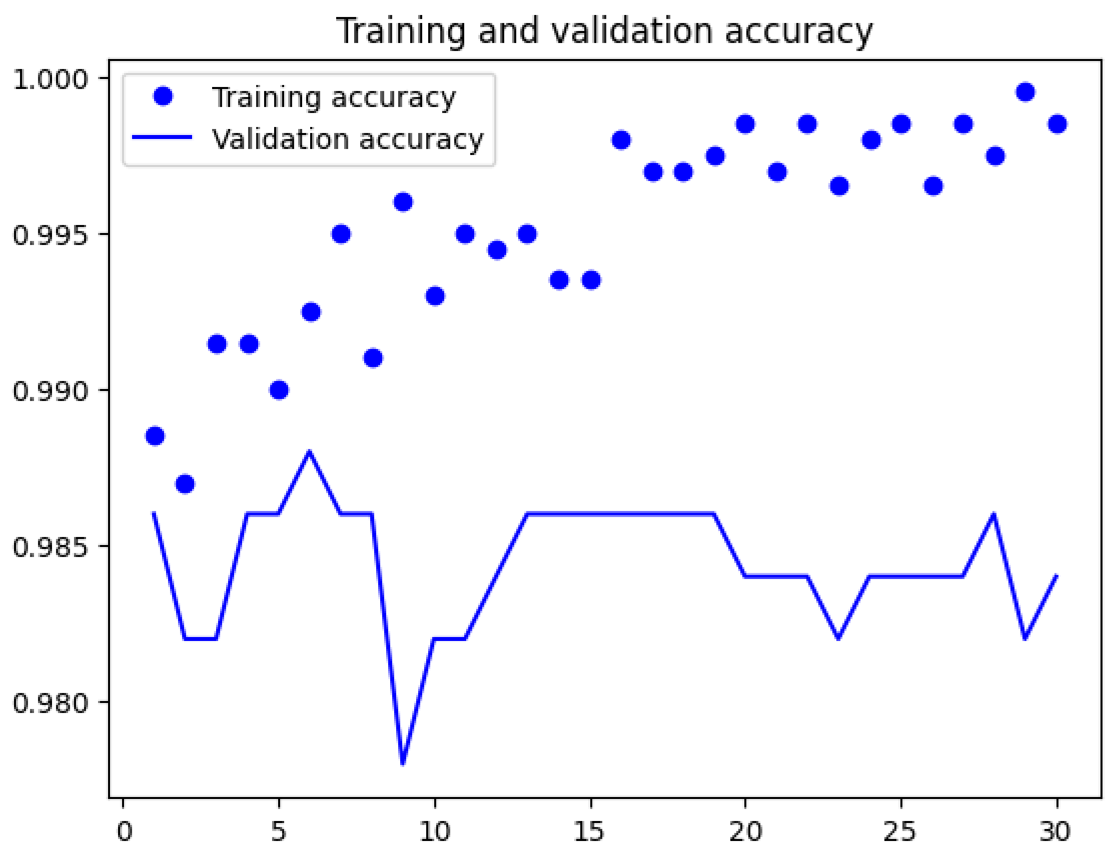
In this figure you can see my baseline model from scratch after 30 epochs. The training accuracy gets right around 76% at best and validation accuracy around 70%.




In this figure you can see my scratch model with 2000 images after 30 epochs. The training accuracy reaches around 90% and validation accuracy reaches about 85%.




In this figure it shows my baseline for my pretrained model with 1000 images. The training accuracy is about 98% and the validation accuracy is around 97%.




In this figure you can see my pretrained model with 2000 images. The training accuracy right about at 99% and the validation accuracy right around 98.5%.

**Scratch Network**

When the convolution neural network was trained from scratch with 1000, training sample images the test accuracy achieved was 70.8%.  After that increasing the training sample size to 1500 which achieved a test accuracy of 75.4%. Finally increasing to a training sample size of 2000 which achieved a test accuracy of 82.8%.

These results show that as the training sample size is increased the scratch model improved in performance but ultimately plateaued at around 83%. This was achieved when increasing the training sample size to 2000 images.

**Pretrained Network**

When the convolution neural network was pretrained with 1000, training sample images the test accuracy achieved was 95.8%.  After that increasing the training sample size to 1500 which achieved a test accuracy of 98.2%. Finally increasing to a training sample size of 2000 which achieved a test accuracy of 97.4%.

The results show that the pretrained model performed better staying around the same accuracy through all tests. The best test accuracy was achieved using 1500 training sample sizes.

## Discussion

The results show an important relationship between training sample size and network choice. When the training sample was relatively small, the pretrained network performed better than the network trained from scratch. This is because the pretrained network has already learned useful visual features from a different large dataset. This makes it more effective when only limited to new training data available. A network learned from scratch must learn all relevant features directly from the current dataset. This makes it more vulnerable to overfitting and lower generalization performance.

As the training sample is increased the performance of the scratch network does start to perform better. With more training images the scratch network can learn more meaningful patterns. This helps with overfitting and improving classification accuracy. The pretrained network still remained best in this case, having the highest test accuracy through every training sample size. This shows that transfer learning is a strong choice for image classification with smaller or moderate sample sizes.

The results suggest that the pretrained network is generally preferable when the dataset is smaller. They provide much stronger performance without the model requiring to learn all visual features from the beginning. The Scratch network becomes more competitive when you can get a larger training sample size.

## Conclusion

In this assignment, convolution neural networks were applied to classify images of cats and dogs using both a pretrained network and one built from scratch. The tests show clearly that training sample size has an impact on the performance of the models.

The network trained from scratch improved as the training sample size got bigger. This shows that more data helps the model learn better features and generalize more effectively. The pretrained network performed better when training data is limited and remained the best accuracy through all training sample sizes demonstrating transfer learning.

Overall, the relationship between training sample size and network choice is that pretrained networks are usually the better option for smaller datasets, while training from scratch becomes more viable when you have more training data available.